Software Requirements Specification (SRS)

ReferrAI – AI-Native Referral & Commission Platform

Version: 2.0

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1. Introduction

Purpose

This document defines the functional and non-functional requirements for ReferrAI, a standalone AI-native referral and commission management platform.

The platform enables businesses to create, manage, optimize, and scale referral, affiliate, and commission-based programs using configurable business rules and intelligent analytics.

Scope

ReferrAI is a configurable referral and commission engine designed to support multiple business models without requiring software customization.

The system provides:

- Referral management
- Affiliate management
- Commission management
- Partner management
- Campaign management
- Fraud detection
- AI-driven insights
- Reporting and analytics

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2. System Overview

ReferrAI is designed as a standalone platform.

The platform shall:

- Manage referral programs
- Manage affiliate programs
- Calculate commissions
- Track referrals

- Process payouts
- Detect fraud
- Generate insights
- Support configurable business rules

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3. Technology Stack

Frontend : React.js
Backend : FastAPI (Python)
Database : PostgreSQL
Authentication: JWT-based Authentication
ORM : SQLAlchemy
Migrations : Alembic
Cache : Redis (Future)
Background Jobs: Celery (Future)
Storage : Object Storage (S3/MinIO)
Deployment : Docker

Notification Services

- Email service
- WhatsApp service

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4. User Roles

Platform Administrator

Permissions:

- Manage platform settings
- Configure business rules
- Configure commission structures
- Manage users
- View analytics
- Manage payouts
- Review withdrawals
- Manage financial reports

Marketing Administrator

Permissions:

- Create campaigns
- Manage promotional assets
- View referral performance

Partner / Affiliate

Permissions:

- Access referral dashboard
- Generate referral links
- View commissions
- Request withdrawals
- Access promotional assets

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5. Functional Requirements

5.1 Authentication Module

FR-AUTH-001

System shall support user registration.

FR-AUTH-002

System shall support login using email and password.

FR-AUTH-003

System shall support password reset.

FR-AUTH-004

System shall support role-based access control.

FR-AUTH-005

System shall support account verification through email.

5.2 Referral Management Module

FR-REF-001

System shall generate unique referral codes.

FR-REF-002

System shall generate unique referral links.

FR-REF-003

System shall generate QR-code-based referral links.

FR-REF-004

System shall track referral clicks.

FR-REF-005

System shall track referral conversions.

FR-REF-006

System shall support configurable referral attribution models.

Supported attribution models:

- Last Click Attribution
- First Click Attribution

Future Support:

- Multi-touch Attribution

5.3 Commission Management Module

FR-COM-001

System shall support percentage-based commissions.

FR-COM-002

System shall support fixed commissions.

FR-COM-003

System shall support recurring commissions.

FR-COM-004

System shall support tier-based commissions.

FR-COM-005

System shall support hybrid commission models.

FR-COM-006

Commission rules shall be configurable through administration settings.

FR-COM-007

Commission calculations shall execute automatically.

5.4 Business Rules Engine

FR-BR-001

System shall provide configurable business rules.

FR-BR-002

Administrators shall configure commission percentages.

FR-BR-003

Administrators shall configure withdrawal thresholds.

FR-BR-004

Administrators shall configure referral eligibility criteria.

FR-BR-005

Administrators shall configure customer incentive programs.

FR-BR-006

Business rules shall be updated without code changes.

5.5 Customer Incentive Module

FR-INC-001

System shall support optional customer incentives.

FR-INC-002

Supported incentive types:

- Percentage discount
- Fixed discount
- Reward credits
- Signup bonuses

FR-INC-003

Customer incentives shall be configurable per campaign.

5.6 Partner Management Module

FR-PRT-001

System shall maintain partner profiles.

FR-PRT-002

System shall support partner tier management.

FR-PRT-003

System shall support automatic tier upgrades.

FR-PRT-004

System shall track partner performance.

5.7 Wallet & Payout Module

FR-WAL-001

System shall maintain partner wallet balances.

FR-WAL-002

System shall support withdrawal requests.

FR-WAL-003

System shall maintain payout history.

FR-WAL-004

System shall maintain transaction history.

FR-WAL-005

Withdrawal approval workflows shall be configurable.

5.8 Campaign Management Module

FR-CAM-001

Administrators shall create referral campaigns.

FR-CAM-002

Administrators shall define campaign incentives.

FR-CAM-003

Administrators shall upload campaign assets.

FR-CAM-004

Administrators shall track campaign performance.

5.9 Marketing Asset Module

FR-MKT-001

System shall support asset uploads.

Supported formats:

- PNG

- JPG
- PDF
- MP4

FR-MKT-002

Partners shall download approved marketing assets.

FR-MKT-003

Assets shall be organized by campaign.

5.10 Fraud Detection Module

FR-FRD-001

System shall detect duplicate accounts.

FR-FRD-002

System shall detect suspicious referral activity.

FR-FRD-003

System shall detect abnormal referral conversion rates.

FR-FRD-004

System shall assign fraud risk scores.

FR-FRD-005

System shall support account freezing.

FR-FRD-006

Administrators shall review flagged accounts.

5.11 AI Intelligence Module

FR-AI-001

System shall analyze referral performance.

FR-AI-002

System shall identify high-performing partners.

FR-AI-003

System shall generate partner performance insights.

FR-AI-004

System shall recommend commission optimizations.

FR-AI-005

System shall recommend campaign improvements.

FR-AI-006

System shall identify fraud patterns.

FR-AI-007

System shall generate referral quality scores.

5.12 Reporting & Analytics Module

FR-RPT-001

System shall provide referral reports.

FR-RPT-002

System shall provide commission reports.

FR-RPT-003

System shall provide payout reports.

FR-RPT-004

System shall provide campaign reports.

FR-RPT-005

System shall provide fraud reports.

FR-RPT-006

Reports shall support export functionality.

Supported Formats:

- CSV
- Excel
- PDF

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6. Database Requirements

Database Technology:

PostgreSQL

Core Entities:

- Users
- Roles
- ReferralProfiles
- ReferralClicks
- ReferralConversions
- Commissions
- Wallets
- Withdrawals
- Campaigns
- MarketingAssets
- BusinessRules
- FraudFlags
- AIInsights
- AuditLogs

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7. Non-Functional Requirements

Performance

NFR-PER-001

API response time shall be under 2 seconds for standard operations.

NFR-PER-002

Dashboard response time shall be under 3 seconds.

Scalability

NFR-SCL-001

Platform shall support future multi-tenant architecture.

NFR-SCL-002

Platform shall support horizontal scaling.

Availability

NFR-AVL-001

System uptime target: 99.5%.

NFR-AVL-002

Daily automated backups shall be maintained.

Security

NFR-SEC-001

Sensitive information shall be encrypted.

NFR-SEC-002

Role-based access control shall be enforced.

NFR-SEC-003

Audit logging shall be maintained.

NFR-SEC-004

HTTPS shall be mandatory.

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8. Competitive Analysis Requirements

The platform shall be benchmarked against:

- Commission Junction (CJ)
- Impact.com
- PartnerStack
- ReferralCandy
- Tapfiliate
- FirstPromoter

The analysis shall identify opportunities for AI-native differentiation.

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9. Future Enhancements

- Multi-tenant SaaS support
- AI-powered campaign automation
- Predictive commission optimization
- Advanced fraud intelligence
- Multi-touch attribution
- Marketplace integrations
- API ecosystem
- White-label deployment

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10. Success Criteria

The platform shall successfully:

- Support configurable referral programs.
- Support configurable commission models.
- Support configurable business rules.
- Operate independently of any specific business.
- Deliver AI-powered referral intelligence.

- Provide scalable referral and commission infrastructure.